

Challenge (Habanero)

- Q1.** Factor the expression $x^2 - 4$
(A) $x^2 - 2^2$
(B) $(x + 2)(x - 2)$
(C) $(x - 2)^2$
(D) $x(x - 4)$
(E) $(x + 1)(x - 1)$
- Q2.** A recipe for a circular cake requires a 12-inch diameter. What is the area of the cake in square inches, given the formula $A = \pi r^2$ and expressing the result as a difference of squares?
(A) 36π
(B) 144π
(C) $\pi(6^2 - 0^2)$
(D) $\pi(12^2 - 6^2)$
(E) $\pi(6 + 6)(6 - 6)$
- Q3.** Factor $25a^2 - 9b^2$
(A) $(5a + 3b)(5a - 3b)$
(B) $(5a - 3b)(5a + 3b)$
(C) $(25a + 9b)(a - b)$
(D) $(25a - 9b)(a + b)$
(E) $a^2 - b^2$
- Q4.** A building's floor plan is 15 meters by 9 meters. The area is expressed as a difference of squares: $A = (a + b)(a - b)$. What is the value of $(a + b)(a - b)$?
(A) $(15 + 9)(15 - 9)$
(B) $(15 - 9)(15 + 9)$
(C) $15 \cdot 9$
(D) $(15 \cdot 9)^2$
(E) $15 + 9$
- Q5.** The height of a right triangle is $3x$ and its base is $4x$. If the area of the triangle is given as a difference of squares
- $A = \frac{1}{2}(a + b)(a - b)$, what is the value of a and b ?
(A) $a = 4x, b = 3x$
(B) $a = 3x, b = 4x$
(C) $a = 2x, b = 5x$
(D) $a = 12x, b = 5x$
(E) $a = 3x, b = x$
- Q6.** A chef has two mixing bowls with volumes $16x^3$ and $9x^3$. The total volume can be expressed as a difference of squares. Factor $16x^3 - 9x^3$.
(A) $(4x)^2 - (3x)^2$
(B) $x^3(4^2 - 3^2)$
(C) $(4x - 3x)(4x + 3x)$
(D) $x(4^2 - 3^2)$
(E) $x^2(4^2 - 3^2)$
- Q7.** The formula for the area of a trapezoid is $A = \frac{h}{2}(a + b)$, but it can also be expressed as a difference of squares. If the height of a trapezoid is 10 and the bases are 8 and 12, factor the expression for the area as a difference of squares.
(A) $\frac{1}{2}(10)(8 + 12)$
(B) $\frac{1}{2}(10)(12 - 8)$
(C) $\frac{1}{2}(10 + 8)(10 - 8)$
(D) $10(10 + 1)(10 - 1)$
(E) $\frac{1}{2}(20)(2)$
- Q8.** A contractor needs to cover a rectangular area of 49 square meters and a square area of 25 square meters. The total area covered can be expressed as a difference of squares. Factor $49 - 25$.
(A) $(7 - 5)(7 + 5)$
(B) $(7 + 5)(7 - 5)$
(C) $7 \cdot 7 - 5 \cdot 5$
(D) $7 + 5$
(E) $7 - 5$

Open Ended Questions (FRQ)

- Q9.** A construction company needs to excavate a site for a new building. The area to be excavated is 121 square meters and the area where they can dump the soil is 36 square meters. If the area excavated and the area where the soil can be dumped can be expressed as a difference of squares, factor the expression for the total area.
- Q10.** A recipe for a cake calls for $a^2 - b^2$ grams of sugar. If $a = 5x + 2$ and $b = 3x - 1$, find the expression for the amount of sugar needed in terms of x and factor the expression as a difference of squares.

Chef's Tips

- Remind students to simplify their expressions after factoring.
- Encourage students to check their calculations using the formula $a^2 - b^2 = (a + b)(a - b)$.
- Suggest students use the factoring method to solve problems related to difference of squares in real-life scenarios.